

Orbitist Modular Vineyard Robot System — Design Report

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1. Introduction

The Orbitist vineyard robot system is a pair of identical differential-drive robots that support each other in the field. One robot works while the other locates it, follows it, docks with it, and — in a later phase — shares its battery with it. Peer-to-peer docking removes the dependence on fixed charging stations, so the pair can run continuously across large or remote blocks.

The platform rests on four ideas:

- **Modular mechanical architecture.** Each robot carries a universal top-deck mounting area for baskets, sensors, or tools, so missions change without changing the base hardware.
- **Interchangeable roles.** Every robot has a front-mounted active claw and a rear passive docking port. Either unit can be host or follower at any time; roles are assigned in software.
- **Cooperative 4WD mobility.** Two docked robots form one mechanically rigid four-wheel-drive unit with better traction on loose or sloped ground.
- **ROS 2 software stack.** Control, sensing, and inter-robot communication run on ROS 2 over DDS, giving modular nodes and distributed coordination.

The first build uses manually swapped battery packs to keep development simple. Every mechanical and electrical interface is designed from the start to accept an autonomous charging module through the docking connection later.

Robot names: *to be decided* (two identical units).

2. Project Objectives

1. **Autonomous task execution.** Each robot performs core vineyard tasks on its own: moving tools, monitoring crop condition, or transporting small payloads on sloped, muddy, or overgrown row paths.

2. **No fixed charging infrastructure.** One robot docks with and extends the range of another in the field, so the system works across large or remote blocks without a central base.
3. **Modular payloads.** Top-mounted modules (harvest bin, camera rig, precision sprayer) swap without redesigning the chassis or electronics.
4. **Dual-role architecture.** Two identical robots assign roles dynamically. If one is low on charge or faults, the other takes over, adding redundancy.
5. **Cooperative 4WD through docking.** Docked robots move as one rigid unit, with a smooth software-coordinated transition from 2WD to 4WD on steep or uneven ground.
6. **Low cost and accessibility.** Raspberry Pi compute and off-the-shelf parts keep the per-unit cost near \$1,400 (Section 12), about a tenth of a commercial micro-tractor.

2a. Buildability Review: What Changed and Why

The original concept is sound, but the two source documents were written at different scales and several part choices fight each other. The revisions below make the robot buildable by two people with a drill press, a 3D printer, and ~\$1,400 per unit, using parts the DIY outdoor-robot community has already proven (hoverboard/scooter hub motors, e-bike packs, Raspberry Pi 5 + ROS 2 Jazzy, AprilTags).

Area	As written	Problem	Revised
Size	18 × 25–30 cm base plate with 12 in (or 23 in) ATV tires	The wheels are wider than the chassis; no room for a 36 V pack, Pi, and payload	55 cm wide × 75 cm long aluminum-extrusion frame; 10 in wheels
Row spacing	1–1.5 m	That is vine spacing along the row; vineyard aisles are typically 2.4–3 m	Design for a 2.4 m aisle, 55–60 cm robot width leaves margin for canopy
Drive	12 V worm gearmotor or 24 V wheelchair motor + \$75 keyed adapter + ATV rim	Three custom interfaces, encoders to add, 12 V/24 V mismatch with battery	10 in pneumatic brushless hub motors (36 V, 350 W, built-in Hall sensors); no gearbox, adapter, or encoder

Area	As written	Problem	Revised
Motor driver	Cytron MD30C (brushed)	Wrong type for BLDC hub motors; original link was a 1 A TB6612 board	Repurposed hoverboard mainboard with EFERU FOC firmware (\$30–60, drives two motors, reports odometry over UART) or a dual FOC controller (ODrive / VESC)
Battery	"12 V SLA" at \$13 (link was a 10 Ah LiFePO4)	Price is wrong; 12 V is too low for hub motors	One 36 V (10S) e-bike pack, 10–15 Ah LiFePO4 or Li-ion, with BMS; 12 V and 5 V via buck converters
Compute	Raspberry Pi 4	Fine for AprilTags; stereo depth from two Pi cams is a calibration rabbit hole	Pi 5 (8 GB) on Ubuntu 24.04 + ROS 2 Jazzy; AprilTag on a Pi Global Shutter camera; depth via an OAK-D Lite only if obstacle avoidance is needed in phase 2
Coupler	Dagu hobby claw gripping a loop	Plastic claw cannot take towing or 4WD traction loads; a fully rigid joint prevents the pair from following ground contours	Cone-guided pin latch: rigid in yaw, free in pitch (like a pintle hitch). Servo releases; a spring holds it closed
Casters	Two 8 in flat-free casters, elevated	Small casters dig into soft soil and jam on trellis debris	Two 10 in pneumatic swivel casters, or drop casters and run all four wheels as skid-steer on a single robot (Amiga-style) — see Section 5

Area	As written	Problem	Revised
Charging	Follower "recharges" the worker through the dock	Two identical packs transferring charge is zero-sum and needs a bidirectional charger	Phase 1: manual swap. Phase 2: docked packs parallel onto one bus through ideal-diode controllers so the pair shares range; the follower can carry a second pack as a power tender
Final-approach sensor	Ultrasonics	Poor returns off foliage and angled plates	Keep AprilTag until contact; a \$5 VL53L1X time-of-flight sensor and a limit switch in the receptacle confirm capture
Software	ROS 1 Noetic scripts	Contradicts the ROS 2 design; Noetic is end-of-life	ROS 2 Jazzy, <code>ros2_control</code> diff-drive controller, <code>apriltag_ros</code> , <code>camera_ros</code>
Cost	\$861	Omits battery, frame, converters, radio, weatherproofing	~\$1,350 per robot (Section 12)

Two things stay exactly as conceived: the symmetric front-active / rear-passive docking geometry, and AprilTag-led docking on Raspberry Pi. Both are the right calls and are already supported by maintained ROS 2 packages.

Reference designs worth borrowing from: [Acorn](#) (Twisted Fields; open-source aluminum ag rover on a Pi CM4, four modular corner drive units), the [farm-ng Amiga](#) (commercial 4WD micro-tractor, e-bike battery packs, ~\$13k — the benchmark this design undercuts), and the [hoverboard-robot](#) / [hoverboard-driver](#) projects (cheapest proven ROS 2 diff-drive base).

3. System Overview

Each robot is a compact differential-drive platform built for structured but imperfect terrain: vineyard aisles of 2.4–3 m, uneven ground, and slopes up to $\sim 15^\circ$. The layout, from front to back, is: active docking latch → front caster → drive-wheel axle with a hub motor either side → battery and compute bay → rear caster → passive docking pin with AprilTag.

Element	Specification
Drive	Differential, two brushless hub motors
Drive wheels	10 in pneumatic scooter/wheelbarrow hub motors, 36 V 350 W, Hall sensors
Casters	Two 10 in pneumatic swivel casters, front and rear
Frame	20×20 or 30×30 mm aluminum extrusion, ~ 55 cm wide × 75 cm long; deck 30–35 cm above ground
Ground clearance	≥ 15 cm under the frame
Front interface	Cone-guided pin latch, servo release, spring closed
Rear interface	Hardened steel cross-pin in a funnel; AprilTag (36h11, 10 cm) above it
Top deck	Flat extrusion rails with T-slot mounting; 12 V and 5 V payload connectors
Compute	Raspberry Pi 5 (8 GB), Ubuntu 24.04, ROS 2 Jazzy
Power	36 V e-bike pack, 10–15 Ah, swappable; 12 V / 5 V buck converters
Target mass	25–30 kg without payload; 20 kg payload
Software	ROS 2 Jazzy over DDS, Wi-Fi mesh between robots

An aluminum-extrusion frame replaces the original flat base plate. It is faster to build (no machining, only cut-to-length and corner brackets), easier to change, and gives T-slot mounting everywhere for the payload and sensors. A thin aluminum or HDPE belly pan closes the bottom against mud.

Key structural constraints:

- Compact enough to pass between rows.

- Enough ground clearance to avoid entanglement with vines and trellis wire.
- Weight distributed so the drive wheels keep traction without overloading the motors.
- Easy access for maintenance, battery swaps, and servo servicing.

The original top-view and front-view sketches from the design document should be inserted here.

4. Chassis, Materials, and Payload Mounting

The frame is aluminum T-slot extrusion (Section 5) with stainless fasteners; sheet material is needed only for the axle clamp plates, belly pan, and payload tray. For those plates, 3 mm aluminum is preferred; the alternatives are ranked below.

Base plate material options

Material	Pros	Cons	Verdict
Aluminum sheet, 2–3 mm	Light, corrosion-resistant, easy to drill	Higher cost than steel	Selected
Aluminum angle / flat bar	Good for frame rails and mounts	Needs assembly	Use with the sheet for rails
Steel sheet, 1.5–2 mm	Very strong and rigid	Heavy; rusts if untreated	Backup
Galvanized sheet	Cheap, pre-treated	Harder to machine	Backup
Plywood	Cheap and strong	Heavy, not waterproof	Prototype only
Acrylic, 3–5 mm	Easy to laser-cut	Brittle under load	Not recommended
3D-printed PLA / PETG	Fully custom geometry	Slow to print; PLA softens in sun	Brackets and small parts only

Payload mounting

Two approaches were considered. **Option A — a removable basket bolted to the chassis — is selected** for phase 1.

	Option A: Basket on chassis	Option B: Chassis is the basket
Modularity	Swap basket for sprayer, sensors, tools	Fixed
Chassis	Flat, simple base frame	Integrated, fewer parts
Prototyping	3D print, laser-cut, or repurpose a container	Requires custom fabrication
Maintenance	Remove basket for instant access to internals	Harder access
Weight / CG	Extra mount structure	Lighter; payload sits lower, better stability

Mounting for option A: bolt-on through a standard hole pattern, or Velcro straps for quick attach/detach during testing. Power and data connectors for payloads are routed to the top deck.

5. Drive System: Hub Motors, Controller, and Casters

Selected: two 10 in pneumatic brushless hub motors (36 V, 350 W, gearless, Hall-sensored) driven by a hoverboard mainboard running EFERU's FOC firmware. This is the cheapest drive train that is already proven under ROS 2 on outdoor robots, and it removes every custom mechanical interface from the original design: no gearbox, no keyed adapter, no separate encoder, no ATV rim.

Why hub motors

- **Torque and speed match the job.** A 350 W gearless hub motor at 36 V delivers roughly 30 N·m and a no-load speed of ~600 rpm; on a 10 in (25 cm) wheel that is a top speed near 8 m/s, so the controller will run it at 10–20 % of range. A 30 kg robot with 20 kg payload climbing 15° needs about 65 N of tractive force, or 8 N·m at the axle — well inside the motor's rating with margin for mud. [Source: typical 10 in 36 V 350 W hub spec](#)
- **Odometry is built in.** Hall sensors give ~90 counts per revolution, enough for the diff-drive controller; the hoverboard firmware reports wheel speed and position over UART.
- **Cost.** \$60–140 per motor with tire, versus ~\$225 per side for gearmotor + adapter + ATV wheel in the original list.

- **Field repair.** The whole wheel is one bolt-off unit; a spare costs less than the original adapter alone.

Drive options compared

Option	Price (2 wheels + control)	Pros	Cons	Verdict
10 in pneumatic hub motors + hoverboard mainboard (EFeru FOC firmware)	\$200–320	Cheapest, proven ROS 2 driver, odometry over serial, 2 motors per board	Firmware flash needs an ST-Link; 6.5 in hoverboard tires are too small — buy 10 in motors separately	Selected
10 in hub motors + ODrive S1 / VESC 6 (one per motor)	\$450–600	Best control quality, CAN bus, well-documented	3× the controller cost	Upgrade path
10 in hub motors + RioRand ZS-X11H boards	\$260–360	Simple PWM/analog interface, documented conversion	No native odometry; needs an MCU in between	Fallback
Original: 24 V wheelchair gearmotor + keyed adapter + ATV rim	\$525+	High torque	Three custom interfaces, encoder to add, voltage mismatch, heaviest	Dropped

Casters, and the skid-steer alternative

Casters are the weak point of any diff-drive robot on soil: small ones dig in, jam on prunings, and fight the drive wheels in turns. Two mitigations, in order of preference:

1. **10 in pneumatic swivel casters** (\$40–60 each, hand-truck style) mounted on a short spring or rubber-block suspension so they load lightly and follow contours.
2. **Skid-steer instead** — four hub motors, two mainboards, no casters. This is what farming's Amiga and most small ag rovers use. It costs ~\$200 more per robot, turns by scrubbing (fine on soil, hard on turf), and makes docking geometry simpler because

there is no caster to collide with the partner's latch. Worth prototyping on robot #2 once robot #1 works.

Frame

20×20 mm (or 30×30 mm for the axle rails) aluminum T-slot extrusion, ~55 × 75 cm, with the hub-motor axles clamped in slotted aluminum plates so track width and axle height can be adjusted. Extrusion needs no machining and every later payload bolts to the same slots.

6. Docking and Coupling Mechanism

Selected: a cone-guided pin latch. The rear of each robot carries a hardened steel cross-pin (10 mm) inside a 3D-printed or sheet-metal funnel. The front of each robot carries a spring-loaded latch jaw that rides up over the pin and snaps closed; a hobby servo (or a small linear actuator) lifts the jaw to release. This is the same principle as a trailer pintle hook or a train coupler, scaled down, and it can be built from a \$12 surface latch plus printed guides.

The joint is deliberately **rigid in yaw and free in pitch** (the pin is horizontal, so the latch can rotate about it). Yaw rigidity is what makes the pair drive straight as one 4WD unit; pitch freedom lets the two robots follow ground contours without lifting a wheel. A fully rigid joint, as in the original design, would either unload a drive wheel on every bump or bend the coupler. Roll compliance comes from the pneumatic tires. Because the pair turns by skid-steer when docked, the software limits docked turning to gentle arcs and headland turns.

The Dagu hobby claw from the original design is kept only as a bench-test actuator: its plastic jaws cannot carry towing or traction loads.

Coupling methods evaluated

Method	Price	How it works	Strengths	Weaknesses	Verdict
Cone-guided pin latch (spring closed, servo release)	\$20–40 built	Latch jaw snaps over a horizontal pin in a funnel	Self-aligning, rigid in yaw, free in pitch, takes towing loads, passive capture	Needs a second actuator for release; custom printed parts	Selected

Method	Price	How it works	Strengths	Weaknesses	Verdict
Spring-loaded hook + latch (train-style)	\$11–14	Hook on A clicks into latch on B	Firm grip, resists push and pull	Fully rigid; needs alignment guide	Donor hardware for the pin latch
U-slot + locking pin (tongue and groove)	\$44	Tongue slides into slot, pin locks	Very stable; easy electrical pass-through	Fully rigid; needs precise approach	Phase 2 if pass-through contacts are added
Servo hobby claw + loop (Dagu Mk II)	\$25	Claw grips a loop or post	Cheap, quick to prototype	Plastic; fails under traction load	Bench test only
Barbed lock	3D-printed	Barbed pin into sprung socket	Self-retaining	Hard to release; wear	Not pursued
RC trailer tow hitch	\$6	Pin joint with play	Cheap	Free in yaw; not rigid enough for 4WD	Rejected
Magnets + guide pins	—	Magnets hold, pins align	No actuator	Separates under vibration	Rejected

Docking station styles considered

- **Hook and latch** (front hook, rear latch): easy to build but demands high approach precision.
- **Mechanical dock + electrical contacts**, as on the [TurtleBot docking station](#) and commercial [AGV charging bays](#): the reference model for the phase-2 charging interface.
- **Magnetic docking**: rejected; breaks apart too easily in the field.

Docking receptacle

The rear cross-pin sits 25 cm above ground, centered, inside a funnel that accepts ± 8 cm lateral and $\pm 10^\circ$ yaw error at first contact. A limit switch behind the pin closes when the latch is seated. The 10 cm AprilTag (family 36h11) is mounted directly above the funnel so the docking camera sees it until the last 20 cm. All parts unbolt from the extrusion for replacement after wear.

7. Electrical System and Compute

One 36 V bus feeds the motors directly; everything else hangs off two buck converters. The whole electrical system fits in one IP65 polycarbonate box (~30 × 20 × 15 cm) bolted to the frame, with the battery in a separate slide-in bay.

Power

Item	Spec	Est. price
Battery	36 V (10S) e-bike pack, 10–15 Ah, LiFePO4 preferred (safer, 2,000+ cycles) or Li-ion NMC (lighter); integrated BMS with 30 A discharge; XT60 or Anderson connector; two packs per pair so one charges while one runs	\$120–220 each
12 V rail	36→12 V buck, 10 A, for payloads, radio, lights	\$15
5 V rail	36→5 V buck, 5 A (Pi 5 needs 5 V / 5 A over USB-C PD or the 5 V pins)	\$15
Protection	30 A main fuse, per-rail fuses, precharge resistor on the motor controller, reverse-polarity MOSFET	\$20
E-stop	22 mm mushroom switch on the deck, cutting the 36 V motor feed via a contactor; plus a wireless e-stop through the RC receiver failsafe	\$40

Runtime estimate: hub motors average ~60 W each on flat soil at 1 m/s, the Pi and sensors ~15 W; a 36 V 12 Ah pack (430 Wh) gives roughly 3 hours of driving. farm-ng quotes 3–8 h for the Amiga on the same e-bike-pack approach. [Source](#)

Compute

Board	Price	Role
Raspberry Pi 5, 8 GB, active cooler, NVMe HAT	\$110	Ubuntu 24.04 + ROS 2 Jazzy: <code>ros2_control</code> diff-drive, <code>apriltag_ros</code> , docking state machine, telemetry

Board	Price	Role
Hoverboard mainboard (EFeru FOC firmware)	\$30–60	Motor FOC, Hall odometry, UART to the Pi
RC receiver (e.g. FrSky / ELRS)	\$25	Manual teleop and hardware failsafe stop
NVIDIA Jetson Orin Nano (phase 2, host robot only)	\$250	Depth and learned perception if the OAK-D route is not enough

Cameras and sensors

- **Docking camera:** Raspberry Pi Global Shutter camera (IMX296, \$50) with a 6 mm lens on the front, looking slightly down. Global shutter matters on a vibrating chassis; rolling-shutter Pi cams smear AprilTags at speed. Detection through `camera_ros` + `apriltag_ros` runs at 15–30 fps on a Pi 5 at 720p. [Source](#)
- **Rear tag:** 10 cm 36h11 AprilTag printed on aluminum composite, above the docking pin.
- **Final approach:** VL53L1X time-of-flight sensor (\$5, 4 m range, 1 mm resolution) on the latch, plus a limit switch in the receptacle. These replace ultrasonics, which return poorly from foliage and angled surfaces.
- **Depth (phase 2):** one Luxonis OAK-D Lite (\$150) on the host for obstacle avoidance and row following; it does its stereo on-camera, so the Pi is not loaded. A DIY stereo pair from two Pi cams is not recommended — calibration drift on a vibrating frame will consume weeks.
- **IMU:** BNO085 (\$20) for heading during docked 4WD motion.
- **Reserved I/O:** battery voltage/current sense (INA226), payload connector (12 V + USB), LoRa or Wi-Fi mesh node.

Servo control

The latch release servo (20 kg·cm metal-gear, \$15) runs from the 5 V rail through a PCA9685 PWM HAT (\$15), never from the Pi's own header, to keep current spikes off the compute supply.

8. Locomotion and Motion Control

Differential drive gives tight maneuvering in narrow rows and enough torque for hauling and docking. The short wheelbase allows in-place turns by counter-rotating the wheels, which is what reversing into docking alignment requires.

Velocity control. `ros2_control` with `diff_drive_controller` takes `/cmd_vel`, computes wheel speeds, and publishes odometry from the hub-motor Hall counts reported over UART. The FOC firmware on the hoverboard board already closes a speed loop per wheel, so the Pi only tunes the outer loop. Wheel-slip compensation comes from fusing odometry with the BNO085 IMU through `robot_localization`.

Cooperative 4WD mode. When two robots are docked they are locked in a straight-line configuration. The host robot's controller takes over both units:

- Motion commands are mirrored to the follower over ROS 2 / DDS on a Wi-Fi link between the two Pis (a small travel router or Wi-Fi mesh node on each robot; LoRa for telemetry only).
- Both robots command the same left and right wheel speeds, so the pair behaves as a four-wheel skid-steer with a ~1.4 m wheelbase. Turning while docked is limited to gentle arcs; sharp turns are done undocked.
- A watchdog on each robot stops its motors if the mirrored command stream drops for more than 200 ms.
- Future work: torque balancing between the two units from measured motor current (Section 15).

9. Perception and Autonomy

The perception stack combines three cheap, proven sensors in a layered approach that runs on low-to-mid range compute.

Range	Sensor	Provides
Long (row-scale, to ~6 m)	AprilTag via Pi Global Shutter camera	Partner ID, heading, lateral offset, full 6-DoF pose
Mid (phase 2)	OAK-D Lite stereo	Depth, obstacle avoidance, row-following, terrain estimate

Range	Sensor	Provides
Short (last 20 cm)	VL53L1X time-of-flight + limit switch	Final distance and confirmed capture; works when the tag leaves the camera's view

AprilTags are the primary fiducial. A tag on each robot's rear docking port lets the follower localize and align during approach. They detect reliably under variable lighting, give accurate pose, run cheaply on a Raspberry Pi, and can encode robot IDs.

Depth is optional for docking. A 10 cm 36h11 tag gives a pose good to a few millimeters at 1 m, which is tighter than the funnel needs. Stereo is added in phase 2 for obstacle avoidance and for navigating to the partner robot when it is out of tag range, using the OAK-D Lite's on-camera depth so the Pi is not loaded.

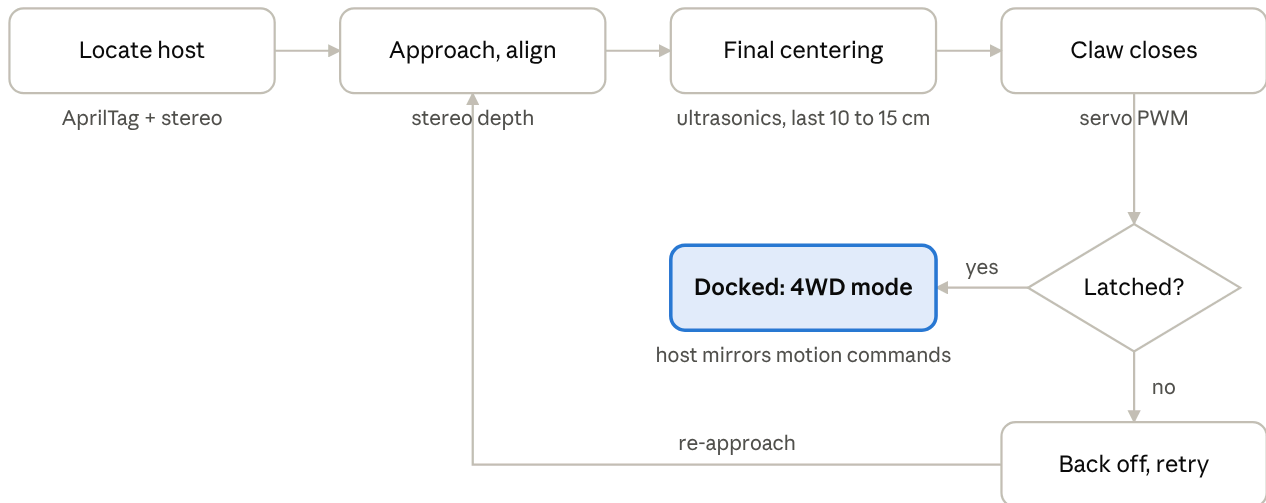
Time-of-flight and the limit switch replace the ultrasonic sensors of the original design: ultrasonics return weakly from leaves and angled plates and have 1–2 cm resolution, while a VL53L1X reads to the millimeter and confirms the last 20 cm when the tag is too close to see.

The full phase-1 pipeline (AprilTag + ToF + limit switch) runs on the Pi 5, so a follower-only robot needs no extra compute. Docking precision targets are set by the funnel: ± 8 cm lateral and $\pm 10^\circ$ yaw at first contact, which the tag pose meets at 1 m with margin.

10. Docking Process and Detection

The follower tracks the host's AprilTag until the last 20 cm, then relies on the funnel, a time-of-flight sensor, and a limit switch to confirm capture before the pair moves as one.

Docking runs through three sensor ranges, then latches into 4WD



docking sequence · 4 steps, 1 decision, retry loop

A failed latch backs the follower off and returns it to the approach phase; a confirmed latch switches the host controller into 4WD mode (and, in phase 2, starts charging).

Latch confirmation

Docking is judged successful when several signals agree:

- The limit switch behind the receptacle pin closes.
- The VL53L1X reads the expected seated distance (± 3 mm).
- The release servo is confirmed at its "closed" position.
- A ROS 2 topic publishes `docked = true`, which gates the 4WD controller; any one signal dropping triggers a stop.

Role selection

Both robots are symmetric, so host and follower are chosen in software from battery level, task status, or an external command. Roles can reverse at any time without hardware changes.

Charging interface (phase 2)

Two identical packs cannot usefully "charge" each other — net energy is conserved and a bidirectional charger would cost more than a third battery. The realistic phase-2 goal is **shared range, not charging**: spring-loaded pogo contacts on the latch and receptacle join the two 36 V buses through an ideal-diode controller (LM5050-class) on each side, so the docked pair draws from whichever pack is fuller and the working robot keeps going on the follower's energy. The follower can also carry a second pack in its payload bay as a power tender. A ROS 2 handshake enables the contacts only after `docked = true`, and current limiting and reverse-polarity protection sit on both sides of the joint.

Open question: does the pin-in-funnel latch capture reliably when the follower approaches on a 10° side slope? This is the first thing to test on the prototype pair.

11. Deployment Use Case

The pair is built for vineyard aisles of 2.4–3 m with sloped, rutted, or muddy ground and prunings underfoot — conditions that stop a single small diff-drive robot.

Primary scenario

Robot A works: hauling picking lugs, carrying a camera rig for canopy or cluster imaging, or towing a small sensor sled. Robot B waits at the row end in standby, listening to A's telemetry (battery voltage, position, task state) over the Wi-Fi mesh.

When A's pack drops below a threshold, or A reports wheel slip on a slope, B switches to follower mode:

1. Drives the row toward A's last reported position (odometry + IMU; GPS-RTK in phase 2).
2. Acquires A's rear AprilTag at ~6 m and closes in on the tag pose.
3. Captures the pin in the funnel; the limit switch and ToF confirm.
4. Publishes `docked = true`; A's controller takes over both drive trains.
5. The pair continues as a 4WD unit; in phase 2 the packs share a bus.

Role rotation

Roles are chosen from battery level and task state, exchanged continuously over telemetry. A robot low on charge requests a follower; an idle robot volunteers; roles reverse when the pair returns to the row end for a pack swap. A base station (a laptop on

Wi-Fi) shows both robots' state and can override roles.

What the pair does that one robot cannot

- Climbs slopes where a 2WD robot with casters loses traction.
- Keeps a job running past one pack's range.
- Hauls a loaded lug tray (20 kg) uphill by sharing the load across four driven wheels.

Scaling

More units add nothing new to the hardware. A fleet scheduler assigns host/follower pairs by battery and proximity; convoys of three or more use the same latch.

12. Bill of Materials (one robot)

Revised BOM for the buildable configuration. Prices are approximate US retail as of September 2026; the original \$861 estimate omitted the battery, frame, converters, radio, and enclosure.

#	Item	Qty	Est. cost (USD)
1	10 in pneumatic brushless hub motor, 36 V 350 W, with tire	2	240
2	Hoverboard mainboard (donor board), flashed with EFeru FOC firmware	1	45
3	ST-Link V2 programmer (one per project)	1	10
4	10 in pneumatic swivel casters	2	100
5	Axle clamp plates, aluminum, water-jet or hand cut	2	40
6	36 V 12 Ah LiFePO4 e-bike pack with BMS	1	180
7	Buck converters, 36→12 V 10 A and 36→5 V 5 A	2	30
8	Fuses, contactor, precharge, wiring, XT60/Anderson connectors	—	60
9	E-stop switch, 22 mm mushroom	1	15

#	Item	Qty	Est. cost (USD)
10	Raspberry Pi 5 8 GB + active cooler + 128 GB NVMe + HAT	1	130
11	Raspberry Pi Global Shutter camera + 6 mm lens	1	65
12	VL53L1X ToF sensor, limit switch, BNO085 IMU, INA226	—	45
13	PCA9685 PWM HAT + 20 kg·cm metal-gear servo	1	30
14	ELRS/FrSky RC receiver (transmitter shared)	1	25
15	Wi-Fi mesh node / travel router	1	40
16	Aluminum T-slot extrusion, brackets, fasteners	—	150
17	IP65 enclosure, glands, belly pan	—	60
18	Docking latch hardware, steel pin, printed funnel, pogo contacts (phase 2)	—	40
19	AprilTags on aluminum composite	2	15
20	Payload tray / lug basket	1	30
	Total		≈ 1,350

Pair cost ≈ \$2,700, plus ~\$150 for an OAK-D Lite on the host in phase 2. A spare drive wheel is \$120; a spare mainboard \$45.

13. Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Latch fails to capture on side slopes or with mud in the funnel	High	Docking aborts	Wide funnel (± 8 cm), open-bottom funnel that sheds mud, retry loop with a back-off, bench-test on a tilt table before field trials

Risk	Likelihood	Impact	Mitigation
Casters dig in or jam on prunings	High	Robot stuck	10 in pneumatic casters on light suspension; skid-steer variant on robot #2 as the fallback
Hoverboard mainboard variant not supported by the FOC firmware	Medium	Drive delay	Buy a board listed as supported in the EFeru firmware wiki before committing; ZS-X11H boards as a fallback
Rolling-shutter smear kills AprilTag detection while moving	Medium	Slow, jerky docking	Global-shutter camera from the start; slow to 0.2 m/s in the last meter
Wi-Fi link drops between robots mid-4WD	Medium	Runaway or stall	200 ms watchdog stops both units; RC failsafe as a hardware backstop
Water and dust in electronics	Medium	Board failure	IP65 enclosure, glanded cables, conformal coating on the mainboard
Pair turns poorly while docked (long skid-steer)	Certain	Reduced maneuverability	Design the workflow so docking happens in the row and turns happen undocked at headlands
Battery handling (36 V Li-ion)	Low	Fire, injury	LiFePO4 chemistry, BMS packs only, fused leads, charge outdoors on a metal tray
ROS 2 learning curve for the team	Medium	Schedule	Start from the hoverboard-robot reference stack rather than from scratch; simulate in Gazebo with the URDF first

14. Testing Plan and Results to Date

Results to date

A ROS 1 Noetic simulation drove a single robot forward for 10 s at 0.05 m/s in Gazebo (`simple_docker` node), and an untested AprilTag pursuit node was drafted. Neither has run on hardware. Both scripts must be ported to ROS 2 Jazzy: `rospy` → `rclpy`, `apriltag_ros` message types, and a `ros2_control` diff-drive controller in place of raw `/cmd_vel` publishing.

Test sequence for the revised design

1. **Bench: drive train.** Flash the mainboard, spin both hub motors from a UART command, confirm Hall odometry against a tape measure over 10 m (target < 2 % error).
2. **Bench: latch.** Mount the funnel and latch on a sled; push the pin in by hand from ± 8 cm lateral and $\pm 10^\circ$ yaw; 50 captures, target 100 % capture and 100 % release.
3. **Bench: AprilTag pose.** Tag at 0.5, 1, 2, 4, 6 m; record pose error and detection rate at 720p on the Pi 5; target < 1 cm at 1 m, > 15 fps.
4. **Simulation.** Two-robot Gazebo world with the URDF, the tag, and the docking state machine; run 100 docking attempts with randomized start poses.
5. **Field: single robot.** Teleop in a vineyard row on gravel and on 10° slope; log motor current, slip, and caster behavior.
6. **Field: docking on flat ground,** then on a 10° side slope. Target: 9 of 10 captures from a 3 m start.
7. **Field: docked 4WD.** Climb the same slope loaded with 20 kg; compare to the single robot.

Each stage produces a short log in this document's Sources tab (or a lab notebook) before the next starts.

15. Roadmap

Phase 1 — Two working robots, manual docking (3–4 months). Build both units from the Section 12 BOM. Teleop over RC; ROS 2 diff-drive with odometry; latch tested by hand and by servo. Exit criterion: both robots drive a vineyard row on RC and the latch captures on the bench.

Phase 2 — Autonomous docking and shared power (3 months). AprilTag docking state machine on the Pi 5; docked 4WD control with the watchdog; pogo contacts and ideal-diode bus sharing; OAK-D Lite on the host for obstacle stop. Exit criterion: 9 of 10 field captures from 3 m; docked pair climbs a 10° slope with 20 kg.

Phase 3 — Useful work (ongoing). Payload modules: lug-tray hauler, canopy imaging rig, soil-sensor sled. GPS-RTK (u-blox ZED-F9P, ~\$250) and row-following with the OAK-D so the follower can find the host out of tag range. Skid-steer variant evaluated on one unit.

Phase 4 — Fleet. Three or more units; base-station scheduler assigns roles by battery and proximity; LoRa telemetry for blocks without Wi-Fi; optional Jetson on the host for learned perception.

Dropped from the original roadmap: bidirectional charging through the dock (replaced by bus sharing) and active suspension (pneumatic tires and pitch-free coupling cover the need at this size).

16. References and Sources

Literature (from the original report)

- Autonomous Alignment and Docking Control for a Self-Reconfigurable Modular Mobile Robotic System. *Robotics*, 13(5), 2024.
- Gil, A., Reinoso, O., & Julián, A. (2012). A Survey on Robotic Technologies for Vineyards. *Journal of Agricultural Engineering Research*.
- Yamamoto, K., et al. (2015). Development of a field robot for autonomous operation in a vineyard. *Biosystems Engineering*, 139, 12–23.

Reference designs and software

- [Acorn precision farming rover — overview](#) (Twisted Fields; open hardware and software)
- [farm-ng Amiga](#) — e-bike battery packs, 3–8 h runtime, ~\$13k platform ([AgFunder News, Jan 2025](#))
- [hoverboard-robot](#) and [hoverboard-driver \(ROS 2 branch\)](#) — cheapest ROS 2 diff-drive base
- [EFeru hoverboard-firmware-hack-FOC](#) — FOC firmware with UART speed/odometry protocol
- [RioRand ZS-X11H conversion](#) — fallback controller
- [apriltag_ros for ROS 2 Jazzy](#) and [AprilTag 3 library](#)

- [WPILib AprilTag introduction](#)
- [TurtleBot docking station](#); [AGV charging bay guide](#)
- [10 in 36 V 350 W hub motor specification](#); [10 in hub motor market prices](#)

Original parts list links (retained for the record): [AndyMark surface latch](#), [cabinet latch](#), [Injora RC hitch](#), [GarveeTech 12 in wheels](#), [iGoPro 23×10.50-12](#), [flat-free casters](#), [worm gearmotor](#), [AGV gearmotor](#), [LiFePO4 pack](#), [Pi camera](#), [Dagu Mk II claw](#).